

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of M. Pharm
(Pharmaceutical Technology and Pharmaceutics (Drug Regulatory Affairs))
University Theory Examination January 2018
PHCC002- BIOPHARMACEUTICS AND PHARMACOKINETICS
Date: 03.01.2018 , Wednesday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** A) What is biotransformation ? Discuss Phase I & Phase-II reactions with examples. **05**
B) Describe absorption model "cell line (Caco – 2) study". Explain blood brain barrier. **05**
- Q.2** A) Derive equation for one compartment open model- I.V. bolus administration. **05**
B) Discuss pharmaceutical factors affecting absorption and distribution of drug. **05**
- OR**
- Q.2** A) Derive equation for two compartment open model-Extra vascular administration. **05**
B) Describe the regulatory aspects of bioavailability and bioequivalence study for 'Controlled release drug delivery systems.'. **05**
- Q.3** **Answer the following (Any three)** **15**
(A) Discuss factors affecting IVIVC.
(B) Discuss patients related factors affecting drug distribution and protein & tissue binding.
(C) Write BCS classification of drugs and discuss its significance.
(D) Derive equation for one compartment open model- I.V. infusion.

SECTION-II

- Q.4** A) Differentiate between: Parallel and Cross-over bio-equivalence study design. **05**
B) Describe urinary excretion study to be performed for the measurement of bioavailability. **05**
- Q.5** A) Discuss different ADME processes that can become saturated resulting in non-linearity. **05**
B) Write a note on: Non compartment analysis. **05**
- OR**
- Q.5** A) Discuss total clearance and volume of distribution in detail. **05**
B) Explain Wagner-Nelson method for estimation of absorption rate constant. **05**
- Q.6** **Answer the following (Any three)** **15**
(A) Explain various pharmacokinetic parameters.
(B) Describe different methods for establishing IVIVC.
(C) Discuss in brief different softwares used for determination of pharmacokinetic parameters.
(D) Write a note on therapeutic drug monitoring.